MD SAIMUN ALAM

Ph.D. Student in Physics | Graduate Research Assistant, Biophotonics & AI Lab
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SUMMARY

- Ph.D. student in Physics developing deep learning methods for label-free optical microscopy and biomedical image analysis.
- Builds end-to-end pipelines for phase-contrast, dark-field and fluorescence data: segmentation, cell tracking, spatiotemporal classification, object detection and virtual staining, delivered as GUI tools for biologists.

EDUCATION

- **The University of Toledo, Toledo, OH, USA** Jan 2025 – Present
  Doctor of Philosophy (Ph.D.) in Physics (expected 2029); Advisor: Dr. Aniruddha Ray
  GPA: 3.93 out of **4.00**
  Graduate coursework: Quantum Mechanics I, Classical Electrodynamics I, Classical Mechanics, Statistical Mechanics, Advanced Atomic & Nuclear Physics Lab, Fundamentals of Modern Physics, Biologically Inspired Computing.

- **Chittagong University of Engineering and Technology (CUET), Bangladesh** 2019 – 2024
  Bachelor of Science in Electrical and Electronic Engineering (EEE)
  CGPA: 3.33 out of **4.00** (**3.63** last semester)

RESEARCH INTERESTS

- Label-free computational microscopy
- Deep learning for biomedical image analysis
- Point-of-care diagnostics

RESEARCH EXPERIENCE

- **Graduate Research Assistant**, Biophotonics & AI Lab, University of Toledo Jan 2025 – Present
  - **Cell death classification:** Developed CytoMorpheus, a label-free deep learning framework that identifies apoptotic and necrotic cancer cells from phase-contrast and dark-field time-lapse microscopy. Designed the imaging experiments, built the segmentation, tracking and classification pipeline, and released it as an open-source desktop application for anticancer drug screening (NIH/NIBIB R15).
  - **Sperm centriole screening:** Co-developed the automated analysis for FRAC (fluorescence-based ratiometric analysis of sperm centrioles), detecting centrioles and axonemes in multichannel microscopy, and developing virtual staining that predicts centriole markers from label-free images, with the Avidor-Reiss Lab.
  - **Modality translation:** Developing an attention-based network that translates between phase-contrast and dark-field images, so a single microscope can provide both contrasts.
  - **On-chip microscopy:** Reconstructing images from lensless on-chip microscopy with microlens arrays, including flat-field correction and automated lens detection, for compact and low-cost imaging.
  - **Point-of-care sensing:** Designed 3D-printed UV imaging enclosures and image-analysis pipelines for a portable fluorescence biosensor that detects microcystin in water; member of the VisioCell handheld microscope team.
  - **Collaborations:** Cell tracking in calcium imaging of kidney cells and protein colocalization analysis for cell biology labs.

- **Undergraduate Thesis Researcher**, Department of EEE, CUET Feb 2023 – Mar 2024
  - **Thesis:** Built a field-collected jute leaf disease dataset and compared CNNs with vision transformers for disease detection (Supervisor: Md. Kamrul Hasan).
  - **Course projects:** VLSI adder design (Cadence), ECG filtering and fuzzy-logic UPFC control (MATLAB).

PUBLICATIONS

Journal Articles

1. M. Jannat, M. S. Uddin, M. A. Hasan, **M. S. Alam**, A. Paul, M. E. H. Chowdhury, J. Haider, "Real-time jute leaf disease classification using an explainable lightweight CNN via a supervised and semi-supervised self-training approach," *Frontiers in Plant Science*, vol. 16, 1647177, 2025. doi:10.3389/fpls.2025.1647177
2. **M. S. Alam**, S. Khoubafarin Doust, A. Ray, "Deep learning assisted label-free microscopy for cell death analysis in real time," *Journal of Biophotonics*, minor revision.
3. L. Achinger, A. Himel, C. Koogan, **M. S. Alam**, S. Taghizadehghahremanloo, S. Sangras, B. R. Harstine, A. Ray, T. Avidor-Reiss, "Automatic FRAC detects low centriole quality in unexplained bull subfertility," in preparation.

Conference Proceedings and Presentations

4. **M. S. Alam**, S. Khoubafarin, A. Ray, "Deep learning assisted label-free microscopy for morphodynamics based rapid cytotoxicity testing," *Optica Biophotonics Congress (Optical Coherence Tomography)*, paper JT4A.39, 2026.
5. A. Himel, C. Koogan, L. Achinger, S. Taghizadehghahremanloo, **M. S. Alam***, T. Avidor-Reiss, A. Ray, "Deep-learning-based screening of sperm centriole integrity using multimodal optical microscopy," *Proc. SPIE 13839, Optics and Biophotonics in Low-Resource Settings XII*, 1383904, 2026. **[Oral, San Francisco, CA]**
6. **M. S. Alam***, S. Khoubafarin Doust, A. Ray, "A comparative evaluation of neural networks for automated label-free screening of chemotherapeutic drugs," *SPIE Photonics West, Label-free Biomedical Imaging and Sensing (LBIS) 2026*, 13863-69, 2026. **[Poster, San Francisco, CA]**
7. **M. S. Alam***, S. Khoubafarin Doust, A. Ray, "Label-free optical microscopy for real-time classification of cell death mechanisms," *APS Eastern Great Lakes Section Fall Meeting*, 2026. **[Oral, Indianapolis, IN]**
8. C. Koogan, A. Himel, **M. S. Alam***, S. Khoubafarin-Doust, A. Kharel, T. Avidor-Reiss, A. Ray, "Automated image analysis for multimodal optical microscopy," *APS Eastern Great Lakes Section Spring Meeting*, 2025. **[Poster, Dayton, OH]**

** presenting author*

SOFTWARE AND DATASETS

- **CytoMorpheus** (GitHub, MIT license): open-source code and desktop application for label-free cell death classification.
- **Jute Leaf Disease Detection** (Kaggle, doi:10.34740/kaggle/dsv/7921248): 920 field images in 3 classes; 5,850+ views and 730+ downloads; reused in published studies.

TEACHING AND SERVICE

- **Graduate Teaching Assistant**, Department of Physics and Astronomy, University of Toledo — 2025 – Present
  Taught undergraduate physics laboratory sections (PHYS 2075, 2085, 2145); exam grading for PHYS 1750.
- **Peer Reviewer**, *Biomedical Signal Processing and Control* (Elsevier), 2 manuscripts — 2026
- **Student Member**, SPIE – The International Society for Optics and Photonics — 2025 – Present

AWARDS AND ACHIEVEMENTS

- **First Place, Tech419 Pitch Competition**, UToledo Business Incubator (VisioCell team, $10,000) — Nov 2025
- **Entrepreneurial Lead, NSF I-Corps Sites Program**, Univ. of Akron Research Foundation — Apr 2025
- **University Technical Merit Scholarship**, CUET — 2022 – 2024

SKILLS

Programming: Python (PyTorch, OpenCV, scikit-learn, NumPy), MATLAB, C
Deep Learning and Vision: CNNs, vision transformers, CNN–LSTM and transformer sequence models, GANs, U-Net, YOLO, Cellpose, Grad-CAM, ensemble learning
Imaging and Optics: phase-contrast, dark-field and fluorescence time-lapse microscopy (Leica DMi8), Zemax OpticStudio, 3D printing
Tools: Git/GitHub, Gradio and Tkinter GUIs, LaTeX, MS Office
Languages: English (IELTS 7.5), Bengali (native)

INDUSTRIAL TRAINING

- **Trainee Engineer**, BSRM Steels Ltd., Chattogram, Bangladesh — Mar 2023

LEADERSHIP AND ACTIVITIES

- **Vice President**, Association of CUETian Ex-Cadets: organized welfare and social programs — 2019 – 2024
- **Event Management Secretary**, CUET Sports Club: organized indoor and outdoor competitions — 2023 – 2024
- **Executive Member**, Green for Peace: tree-planting (500+ trees) and hygiene campaigns — 2022 – 2023

REFERENCES

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Md. Kamrul Hasan
Relation: B.Sc. Thesis Supervisor
Assistant Professor, Department of Electrical and Electronic Engineering
Chittagong University of Engineering & Technology, Chittagong, Bangladesh-4349
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